CueBridge *RS*

by ideaΔmplifier

User Manual

Connecting between Reaper and SSL Live

Version	**v0.2.8**
Date	**2026-11**

> This manual matches the English interface inside the app; buttons and field names that actually appear on screen are shown in `monospace`.

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1. What this is

CueBridge RS is a bridge between an SSL Live console and REAPER, built for live multitrack recording. It does **no audio processing** itself. It does three things:

Function	What it does
SESSION BUILDER	Turns the SSL channel list (CSV) into a new REAPER project: track names and order match the console, and record inputs and hardware outputs are patched in recorder order
AUTO-MARKING	Places a named marker in the REAPER recording every time an SSL scene fires
SSL DAW CONTROL	The console's own DAW keys (DAW 1 popout, User Keys, Events) drive REAPER record, play, loop and marker jumps over ipMIDI (HUI); REAPER's state feeds back to the console key lights

For: anyone recording live multitracks on an SSL Live console and mixing afterwards with everything lined up to the scenes.

Not for: rigs without SSL Live; situations needing automatic mixing or automatic EQ (this tool does no audio processing at all).

2. System requirements

- **macOS 13 (Ventura) or later** — tested on a Mac running macOS 13.7.1
- Universal build: runs natively on Apple Silicon and Intel
- **REAPER**
- **SSL Live V5.2 or later** (this tool assumes recorder I/O is 1:1 in Console Config order)
- Network: one Ethernet cable, Mac ↔ SSL **Connectivity port**; this subnet **must not share a segment with Dante**
- No ipMIDI driver to install

3. Installation

1. Open `CueBridge-RS-v0.2.8-Mac.pkg` and follow the prompts
2. Open CueBridge RS from Applications

4. One-time setup (five minutes)

4.1 SSL console side

1. **Options › Network**
   - Put the Connectivity port on a subnet **different from Dante**
   - Connect the Mac to the same subnet
2. **Options › DAW**
   - **DAW 1 = HUI**, assigned to an ipMIDI port (example: **2**)
   - Tick **Transport Master** if you want the User Keys to control transport
3. **Scene Output Action**
   - Each scene's MIDI action should Make both **local MIDI** (for Fourier, say) **and ipMIDI 1**
   - DAW Control uses **another** ipMIDI port (example: 2)
4. Notes
   - **The DAW channel carries no audio**; do not put it in a bank
   - If the DAW popout in the Channel View sidebar shows DAW 2 and is all greyed out → **press and hold** to switch back to DAW 1

4.2 REAPER side (once per Mac)

1. **Quit REAPER first** (REAPER rewrites `reaper.ini` on exit)
2. On the **AUTO-MARKING** page in CueBridge, click `Set Up Reaper OSC…` → `Set Up`
3. Reopen REAPER

What it does:

- Adds an OSC control surface to the `[reaper]` section of `reaper.ini`: `CueBridge_RS`
  - REAPER listens on **49121** and replies on **49122**
- Writes a trimmed OSC pattern: `OSC/CueBridge.ReaperOSC`
  - The default pattern computes track names, volume, VU, FX and six time formats on every surface update (about 290 messages/sec while playing); the trimmed one sends about 12/sec
  - This stops REAPER dropping the insert-marker command under load when recording with all 90 tracks armed
- **Backs up** `reaper.ini`: `reaper.ini.before-CueBridgeRS.bak` plus a timestamped backup
- Every other setting is **left untouched**; no MIDI configuration is needed inside REAPER

> Recommended: in REAPER's **Preferences → Control/OSC/web**, delete any OSC surface you are not using.

4.3 CueBridge side

Set these on the **AUTO-MARKING** page:

Field	Setting
`Source` on `SCENES IN`	`ipMIDI 1 · network`; pick the device instead if you go through a USB MIDI interface or IAC
`Network` on `SSL DAW CONTROL`	The network interface connected to the SSL
`DAW port` on `SSL DAW CONTROL`	The same ipMIDI port as **DAW 1** on the SSL (example: 2)

- The IP of the first console to send HUI is locked in and shown on screen as `Console`; every other ipMIDI source on the network is ignored
- **After a venue change or a console IP change**, click `Re-detect` to scan again

5. SESSION BUILDER: building the REAPER project before the show

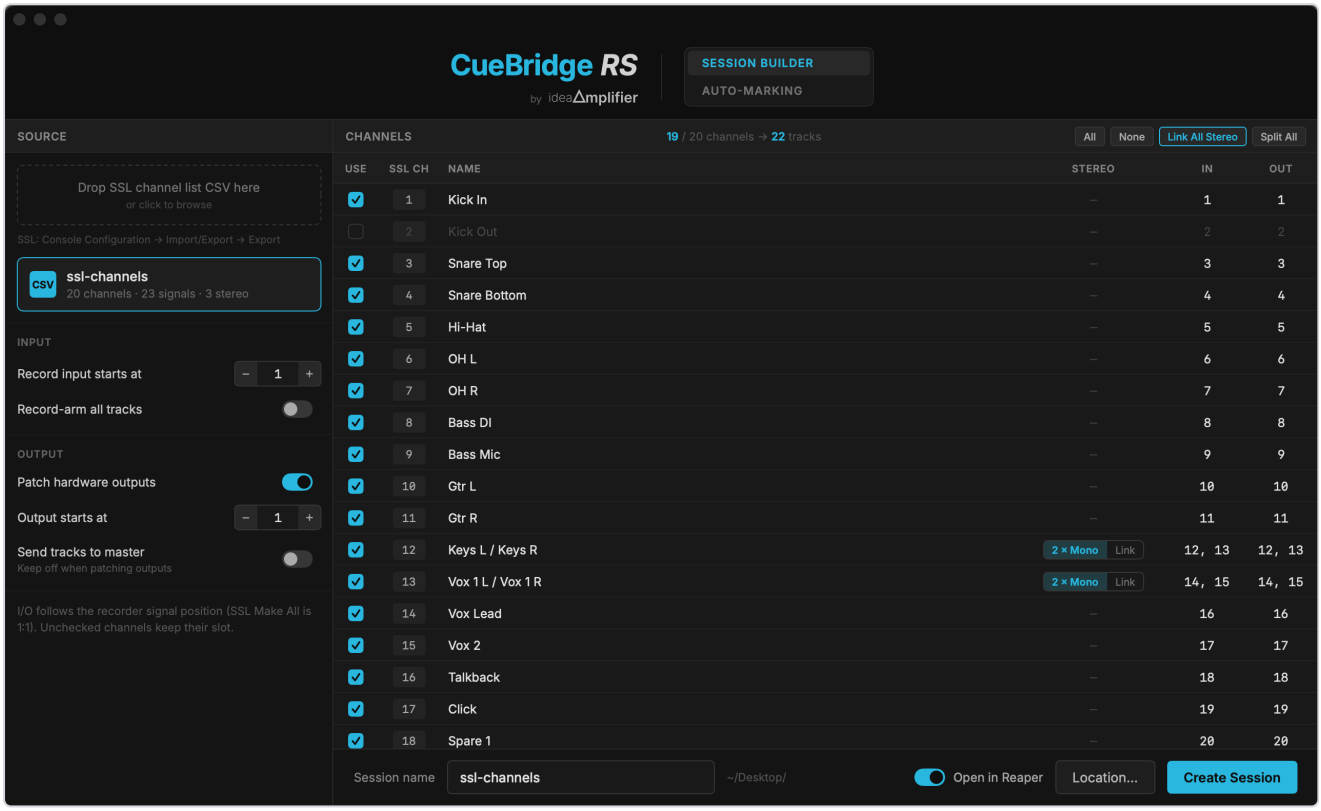


Fig. 1 · SESSION BUILDER: the track table after loading the SSL channel list

5.1 Export the SSL channel list

On the SSL: **Console Configuration → Import/Export → Export** to a CSV.

5.2 Load

Drag the CSV onto `Drop SSL channel list CSV here` in the left column, or click to browse.

After loading, the left column shows the filename and a summary: `N channels · N signals · N stereo`.

5.3 Left column options

Option	Default	Notes
`Record input starts at`	1	REAPER input number for the first recorder signal
`Record-arm all tracks`	Off	Whether every track is armed after the build
`Patch hardware outputs`	**On**	Whether REAPER hardware outputs are written
`Output starts at`	1	First output number
`Send tracks to master`	**Off**	Keep this off when outputs are patched

> **Warnings appear automatically**:
>
> - Signal count > 64 → one MADI recorder has only 64 channels at 48k; check the I/O from the 65th signal onwards
> - Patching outputs and sending to master at the same time → if the master is routed to outputs 1/2, playback feeds the whole mix into channel 1's playback return

5.4 Right column track table

Column	Notes
`USE`	Whether this track is created
`SSL CH`	Console channel number/name
`NAME`	Track name (a linked stereo pair shows a single linked name)
`STEREO`	Stereo channels only: `2 × Mono` or `Link`
`IN`	The corresponding record input number
`OUT`	The corresponding hardware output (`–` when not patched)

- The centre of the header shows `ticked / total channels → N tracks`
- Buttons above: `All` / `None` / `Link All Stereo` / `Split All`
- **One CSV row = one recorder signal**; a stereo pair already occupies two rows in the CSV
- **I/O follows the recorder signal position**: unticking a track does not shift the numbers of the tracks after it

5.5 Create

At the bottom:

- `Session name` — defaults to the CSV filename
- `Location…` — defaults to the Desktop
- `Open in Reaper` — opens it straight after the build (on by default)
- `Create Session`

File created: `<Location>/<Session name>/<Session name>.RPP`

- A `.RPP` of the same name already exists → a dialog appears (`OK` / `Show in Finder`) and it is **not overwritten**
- After a successful build, AUTO-MARKING switches automatically to this show's scene list

6. AUTO-MARKING: placing markers during the show

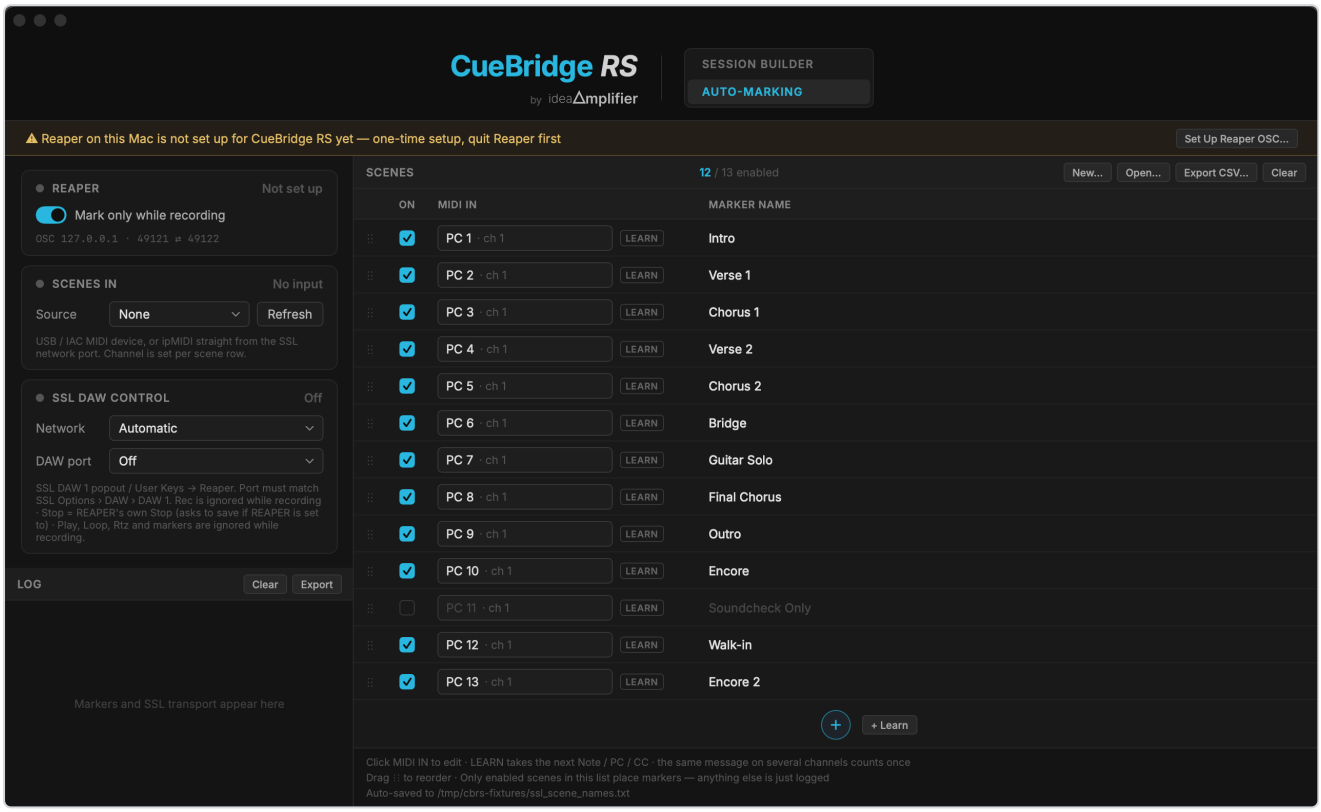


Fig. 2 · AUTO-MARKING: status cards, Log and scene list (the top banner appears when REAPER is not set up; see §4.2)

6.1 The three status cards

REAPER

Status	Meaning
`Not set up`	The OSC setup in §4.2 has not been done
`No response`	Set up, but no reply from REAPER within 8 seconds → click `Retry`, or check that REAPER is running
`Port in use`	49122 is taken (commonly: another CueBridge or OSC tool is also running)
`Connected · Stopped` / `Connected · Playing` / `Recording`	Normal

- `Mark only while recording` (**on by default**): places markers only while REAPER is recording. Turn it off and markers are placed while recording or playing

SCENES IN

Status	Meaning
`No input`	No source selected
`Listening`	Source selected, no message received yet
`Receiving`	Receiving (the status light flashes)
`Disconnected`	The selected source has gone away → click `Refresh`

SSL DAW CONTROL

Status	Meaning
`Off`	`DAW port` = Off
`Waiting for console`	Port selected, no HUI from the console yet
`Online`	Normal
`Network error`	Problem with the network interface or port

6.2 Scene list

The list sets **which scenes get markers, and what names they get**.

Column	Notes
`ON`	Enable/disable this row
`MIDI IN`	The trigger. Click it to open the editor, or use `LEARN` to capture the next MIDI message
`MARKER NAME`	The marker name written into REAPER

MIDI IN file syntax (.txt/.csv)

Syntax	Meaning
`21`	Program Change 21 on channel 1 (the program number shown on the SSL screen, 1–128)
`21@5`	PC 21, channel 5
`21@any`	PC 21, any channel
`CC:20=127@1`	CC 20 = 127, channel 1
`CC:20@any`	CC 20, any value, any channel
`NOTE:60@1`	Note On 60, channel 1 (60 = C3, matching the SSL)
`NOTEOFF:60@1`	Note Off 60, channel 1

Controls

- The round `+` at the bottom: adds a row (PC defaults to the highest value + 1)
- `+ Learn`: captures the next Note/PC/CC into a new row; while waiting the button reads `Waiting for MIDI…`, click again to cancel
- `⋮⋮` drag to reorder; drag into the drop area at the bottom = move to last
- `New…`: opens a separate blank list (the current list is untouched)
- `Open…`: opens a `.txt` list or a CSV produced by `Export CSV…` (the CSV is saved as a `.txt` of the same name in the same folder; the CSV itself is untouched)
- `Export CSV…`: exports `MIDI In,Name,Enabled` for tidying up in a spreadsheet
- `Clear`: empties the current list (saved immediately, cannot be undone)

Rules

- **Only rows in the list that are enabled place markers**; nothing else does
- A message arriving for a disabled row → the Log shows `Skipped · scene disabled`
- A PC outside the list → the Log shows `Skipped · not in scene list`
- Notes/CCs outside the list → not logged
- The same `MIDI IN` on more than one row → ⚠ appears below the list and **only the first enabled row is used**
- SSL's **All Channels** sends the same message on 16 channels (copies arrive about 70–130 ms apart in the field): within 0.5 s, a channel not yet seen counts as a copy and is counted once; the same channel appearing again = a genuine second trigger

Saving

- Saved automatically to **`ssl_scene_names.txt`** next to the session

- Disabled rows are saved as `# trigger,name` (lines starting with `#` are comments; the Windows Lua version reading the same file skips them automatically)
- PC + ch 1 is written as `21,name`, compatible with older versions and the Lua version

6.3 Marking sequence

1. A matching MIDI message arrives
2. Sends REAPER `action 40157` (insert marker at the current position)
3. Waits for REAPER to report a change in `/lastmarker/number/str`
4. Uses `/lastmarker/name` to rename the new marker to the name from the list

- **No confirmation within 2 seconds → no rename** (to avoid renaming some other marker); the Log shows `no reply from Reaper, not named`. This 2 s is the rename-confirmation timeout, a separate mechanism from the 8 s connection keep-alive in §6.1
- `LOG` in the left column lists time, trigger, name and status in order; `Clear` empties it, `Export` writes a CSV

6.4 SSL DAW CONTROL mapping

SSL control	REAPER action
Rec	Start recording
Stop	REAPER's own Stop (whether it asks to save depends on REAPER preferences)
Play	Play
Loop	Loop
Rtz	Return to the start of the project
Prev/Next	Jump to the previous/next marker
Goto Marker 1	Jump to marker 1

Accidental-trigger protection (REAPER's `/record`, `/play` and `/repeat` are toggle commands):

- While recording → `Rec` is ignored
- While recording → `Play`, `Loop`, `Rtz` and marker jumps are ignored
- While playing → `Play` is ignored
- Already stopped → `Stop` is ignored
- While playing or recording → `Stop` always works

The SSL handles **Prev/Next** counting and sends the resulting number over the HUI numeric keys (`. N .`) to REAPER as `/marker N`.

REAPER's play/record/loop state is sent back to the console and lights the keys on the DAW panel.

7. Show-day checklist

Load-in

- ☐ Mac connected to the SSL Connectivity port over **wired Ethernet** (ipMIDI over Wi-Fi drops packets, and it gets worse once the audience is in)
- ☐ The SSL DAW popout shows **DAW 1**
- ☐ CueBridge: `SSL DAW CONTROL` has the right interface and `DAW port`, status `Online`
- ☐ `SCENES IN` status `Listening` (fire a scene to check it goes `Receiving`)
- ☐ Venue changed → click `Re-detect`

Rehearsal

- ☐ Project built in SESSION BUILDER, track count and I/O confirmed correct
- ☐ Scene list loaded, names correct
- ☐ Fire 2–3 scenes as a test, confirm the markers appear with the right names
- ☐ Confirm the `Mark only while recording` setting suits how you work

During the show

- ☐ **AUTO-MARKING** page open with `LOG` visible; `Skipped · not in scene list` means the list is missing a scene
- ☐ If `no reply from Reaper` appears, first check REAPER has no dialog open

Wrap

- ☐ Export the Log CSV with `Export` and keep it on file for reference

8. Troubleshooting

Symptom	Likely cause	Fix
REAPER stuck on `No response`	REAPER is not running, the OSC surface was deleted, or 49121 is taken	Confirm REAPER is running and click `Retry`; if the surface was deleted, quit REAPER, relaunch CueBridge RS, then click `Set Up Reaper OSC…`; check for another OSC tool
`Port in use`	Another CueBridge or OSC program holds 49122	Quit the duplicate program
`Waiting for console` never changes	Wrong interface selected, `DAW port` does not match the SSL's DAW 1, or the two are not on the same subnet	Check §4.1/§4.3; if the screen still shows the old `Console` IP, click `Re-detect`
DAW popout all greyed out	The console is looking at DAW 2 (no port assigned)	Press and hold the DAW popout to switch back to DAW 1
Scene fires but no marker	A scene outside the list, a disabled row, `Mark only while recording` on while nothing is recording, or REAPER is stopped	Read the `Skipped · …` reason in `LOG` (Note/CC outside the list are not logged)
One scene places 16 markers	Behaviour of versions before 0.2.2; the current version dedupes by channel	If it still happens, confirm the installed version is ≥ 0.2.2
Markers dropped with all 90 tracks armed	REAPER load too high	If the setup banner is showing, quit REAPER and click `Set Up Reaper OSC…` again; delete unused OSC surfaces
Nothing at all after a venue change	The console IP is still locked to the previous one	Click `Re-detect`
The setup banner appears again at launch	The OSC setup is missing or from an older version	Quit REAPER → click `Set Up Reaper OSC…` again
The `Console` IP on screen is wrong	Another machine on the network is sending HUI	Only the first source is locked in; confirm no other console or practice application is sending HUI
Installer says it cannot install in this location	macOS version below 13	Check your system version

Field debugging tool: `tools/ipmidi-probe.swift` in the repo can capture ipMIDI/HUI packets.

9. File locations

File	Location
OSC pattern	`~/Library/Application Support/REAPER/OSC/CueBridge.ReaperOSC`
reaper.ini backup	`~/Library/Application Support/REAPER/reaper.ini.before-CueBridgeRS.bak` plus timestamped backups
Scene list	`ssl_scene_names.txt` inside each session folder; with no session, in `~/Library/Application Support/CueBridge RS/`
Log export	Wherever you choose with `Export`

> Only the standard REAPER install location is supported for `reaper.ini`; portable installs are not.

10. Known limitations

- **Listens only to the first console that sends HUI**; other ipMIDI sources on the same segment are ignored
- No automatic updates; new versions must be downloaded and installed by hand
- REAPER portable installs are not supported
- SESSION BUILDER assumes SSL recorder I/O is **1:1 in Console Config order** (Make All on SSL Live V5.2+)
- Scene MIDI triggers must be set up separately in the SSL's Scene Output Action; CueBridge does not change console settings
- Tested on: macOS 13.7.1 (Intel) and macOS 14 (Apple Silicon); SSL Live (SOLSA L550) + REAPER, including a live recording on 2026-10-04

11. Licence and acknowledgements

- © 2026 Idea Amplifier · All rights reserved.

- Designed & built by **YO Hsiangchu Yu**
- Bug reports: ideaamp.support@icloud.com
- More tools: https://github.com/IdeaAmplifier
- This software is **free** and provided as is, without warranty of any kind
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- The ideaAmplifier series: **SpaceRanger** (space), **Baseline** (timing), **CueBridge** (console ↔ recording), **PhatAnchor** (phase)
